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ORIGINAL SCIENTIFIC PAPER

Impact of tocilizumab on endothelial function following out-of-hospital cardiac arrest: a substudy of the IMICA randomized controlled trial

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Aims

Biomarkers of endothelial activation and damage are elevated after out-of-hospital cardiac arrest (OHCA). Elevated plasma concentrations of syndecan-1, soluble thrombomodulin (sTM), and platelet endothelial cell adhesion molecule 1 (PECAM-1) reflect glycocalyx degradation, endothelial cell injury, and endothelial junction disruption, respectively. Interleukin 6 (IL-6) plays an important role in the inflammatory mechanisms that cause endothelial damage, and inhibiting IL-6 with tocilizumab may attenuate endothelial damage. The present study examines the effect of tocilizumab on endothelial biomarkers and their prognostic value in comatose resuscitated OHCA patients.

Methods and results

The ‘IL-6 Inhibition for Modulating Inflammation after Cardiac Arrest’ (IMICA) trial included 80 comatose OHCA patients who were randomized to receive tocilizumab (8 mg/kg) or placebo. Blood samples were drawn sequentially at hospital admission (0 h) and after 24, 48, and 72 h for biomarker measurements. Syndecan-1 concentrations declined over time in both groups; however, the reduction was significantly less in the tocilizumab group (all $P < 0.003$). No significant between-group differences were observed for sTM or PECAM-1. In prognostic analyses, baseline PECAM-1 had an AUROC of 0.71 (95% CI 0.59–0.83) for prediction of 30-day mortality. Syndecan-1 and sTM had no predictive value (AUROC 0.61 and 0.53, respectively).

Conclusion

Plasma syndecan-1 decline was less pronounced in comatose OHCA patients treated with tocilizumab, without affecting sTM or PECAM-1. This contrasts with previous IMICA findings of reduced systemic inflammation, vasopressor use, and myocardial injury, suggesting a complex relationship between inflammation and endothelial injury. Finally, PECAM-1 levels already at hospital admission had a moderate predictive value for mortality.

Clinical trial registration

The trial is registered at clinicaltrials.gov (NCT03863015).

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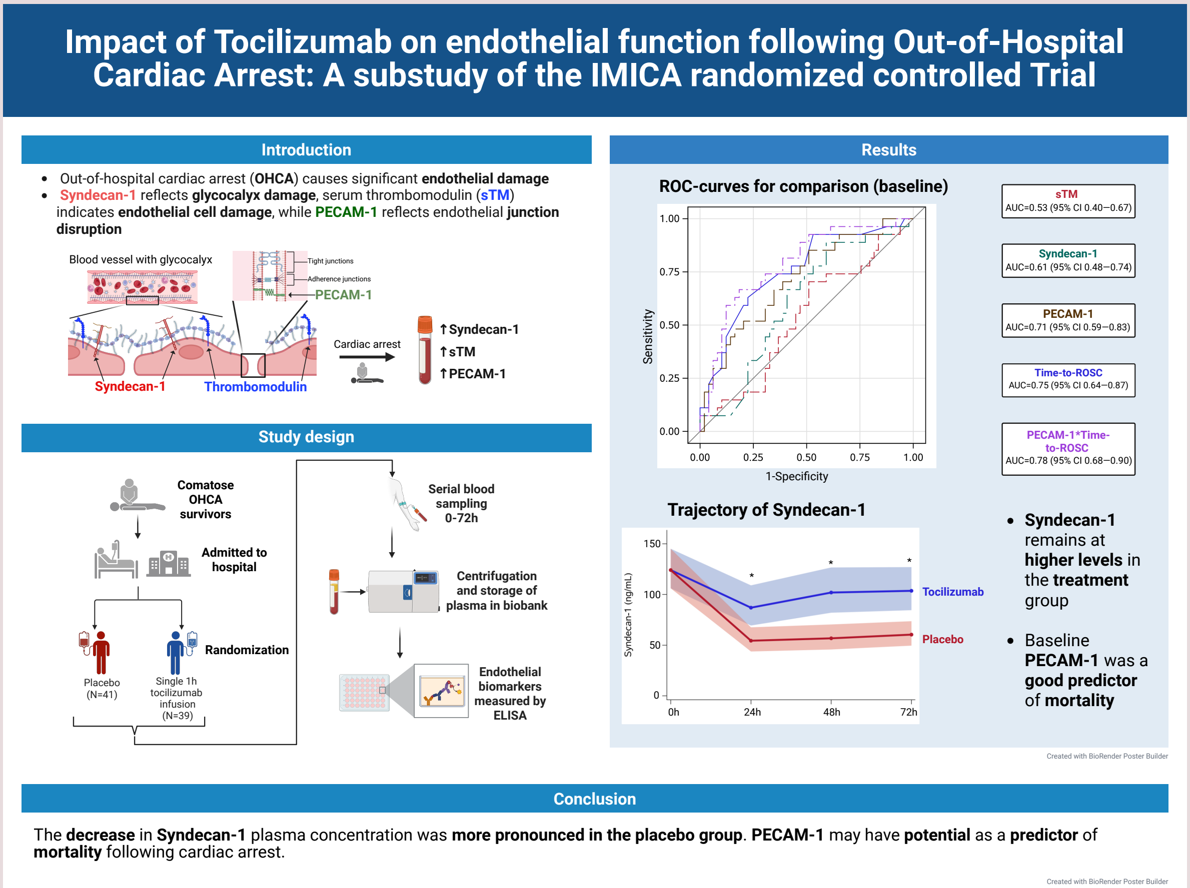
Graphical Abstract



Keywords Syndecan-1 • Soluble Thrombomodulin • Platelet endothelial cell adhesion molecule-1 • Out-of-hospital cardiac arrest • Biomarkers • Tocilizumab • Prognostication

Introduction

Out-of-hospital cardiac arrest (OHCA) represents a major cause of death in Europe, imposing a significant burden on health care systems, patients, and their families.[1] The considerable morbidity and mortality associated with OHCA are partially attributed to the systemic ischaemia–reperfusion response, one of four components in the widely recognized post-cardiac arrest syndrome.[2–4] This ischaemia–reperfusion injury exerts its effects across multiple organ systems, including the functionally significant organ, the endothelium.[5]

The vascular endothelium plays a central role in the pathogenesis of systemic inflammation.[6] During the reoxygenation phase following resuscitation, reperfusion injury induces endothelial damage, partly mediated by rapid increase in reactive oxygen species.[7] This injury is accompanied by elevated circulating markers, indicative of endothelial activation and damage.[6,8–13] Recently, endothelial markers reflecting endothelial glycocalyx damage—namely, syndecan-1 and endothelial cell damage—soluble thrombomodulin (sTM), and endothelial junction disruption, platelet endothelial cell adhesion molecule 1 (PECAM-1), have been associated with adverse outcomes following OHCA.[10,14]

Endothelial activation induces increased vascular permeability, activation of the coagulation cascade, and the release of pro-inflammatory cytokines.[8,15] Among these cytokines, interleukin-6 (IL-6) plays a pivotal role in mediating systemic inflammation, with abundant effects. IL-6 is involved in the initiation of fever and the acute-phase response, enhances vascular permeability, and can contribute to myocardial injury.[16] Additionally, IL-6 can directly activate the endothelial cells, which subsequently release IL-6 and IL-8, leading to a cytokine storm that correlates with the severity of post-cardiac arrest syndrome.[16–18] The released IL-6 can also cause endothelial cell damage, leading to vascular leakage.[16]

Elevated IL-6 levels after OHCA are associated with increased 30-day mortality.[19] Additionally, studies of reperfusion injury following myocardial infarction have shown that IL-6 is linked to the degree of myocardial damage, mortality, and adverse effects.[20–22]

Since IL-6 plays a central role in the pathophysiology of many inflammatory diseases, targeted therapies have been developed to counteract its effects.[23] Tocilizumab, a monoclonal antibody directed at the IL-6 receptor, has been used to treat many rheumatic diseases.[23] Recent findings from the IMICA trial showed that tocilizumab treatment significantly attenuated the inflammatory response seen after cardiac arrest.[24] Additionally, tocilizumab was associated with a reduction in myocardial injury and stress, evidenced by lower troponin-T and NT-proBNP levels.[24] A substudy of the IMICA trial further demonstrated that patients treated with tocilizumab exhibited a more stable cardiovascular function compared with placebo, as indicated by a lower vasoactive–inotropic score at 24 h.[25]

The aim of this study is to evaluate the effect of tocilizumab on endothelial injury markers (syndecan-1, sTM, PECAM-1) over 0–72 h and assess their prognostic value in comatose resuscitated OHCA patients.

Methods

Study design

This is a substudy of the IMICA trial, a randomized, placebo-controlled, double-blinded single-centre study investigating tocilizumab's anti-inflammatory and potential organ-protective effects in 80 comatose resuscitated OHCA patients. Following hospital admission, patients were randomly assigned to receive either a single 1 h infusion of tocilizumab (8 mg/kg, maximum 800 mg) or placebo, in addition to standard care, in a 1:1 fashion. As part of standard care, patients undergoing acute coronary angiography (CAG) received systemic heparinization. Study approval was achieved from the Danish Medicines Agency (Approval ID 2018-002686-19) and the Regional Ethics Committee of the Capital Region of Denmark (ID: H-20022320). The trial was registered at ClinicalTrials.gov (Identifier: NCT03863015). The previously published study protocol provides an in-depth description of the study design, rationale, safety, and statistical analyses.[26] The main findings of the trial have previously been published.[24,25]

Study population

The study population comprised 80 comatose resuscitated OHCA patients admitted to a tertiary heart centre between March and December 2019. Inclusion criteria included OHCA with presumed cardiac causes, age over 18 years, unconsciousness at hospital admission (Glasgow coma scale ≤8), and sustained return of spontaneous circulation (ROSC) for more than 20 min. Detailed inclusion and exclusion criteria are described in the previously published study protocol.[26]

Blood samples

Blood samples were drawn sequentially at hospital admission (0 h) and after 24, 48, and 72 h. Biobank samples were centrifuged for 10 min at 2000 × *g*. Plasma was extracted and stored at −80°C before being thawed for analysis. The following three endothelial markers were measured in citrate (syndecan-1 and sTM) and EDTA (PECAM-1) plasma using different biomarker analysis methods: syndecan-1 [Diaclone, Nordic Biosite, Copenhagen, Denmark; lower limit of detection (LLD) 4.94 ng/mL], sTM (Diaclone, Nordic Biosite, Copenhagen, Denmark; LLD 0.31 ng/mL), and PECAM-1 [R&D Systems Europe, Ltd., Abingdon, UK; minimal detectable dose (MDD) 0.021 ng/mL]. According to the manufacturer, the reference plasma concentration ranges, based on samples from healthy individuals, are 15.7–68.9 ng/mL for syndecan-1, 2.9–7.6 ng/mL for sTM, and 7.18–17.1 ng/mL for PECAM-1.

Study outcomes

The objective of this substudy of the IMICA trial is to determine the effect of tocilizumab on the endothelium, as indicated by changes in syndecan-1, sTM, and PECAM-1 measured serially at 0–72 h. Furthermore, it aims to investigate the prognostic abilities of these markers in comatose resuscitated OHCA patients. Finally, it compares the concentration of the three biomarkers in 30-day survivors vs. non-survivors and syndecan-1 concentrations in patients treated with CRRT vs. no CRRT within the tocilizumab group.

Statistical methods

The statistical analyses were performed using SAS Enterprise Guide 8.4 (SAS Institute Inc.). *P*-values of less than 0.05 were considered statistically significant.

Baseline characteristics were presented using medians and interquartile ranges (IQR) for continuous variables and frequencies (*N*) and percentages (%) for categorical variables.

Endothelial marker data underwent $\log_2$ transformation and were analysed using baseline-corrected repeated measurement mixed models (SAS PROC MIXED). Missing values were handled directly by the mixed model using maximum likelihood estimation. The trajectory of endothelial biomarkers is presented as geometric means with 95% confidence intervals after antilog transformation.

To evaluate the predictive capability of baseline values of syndecan-1, sTM, and PECAM-1 on 30-day mortality, the receiver operating characteristic curve (ROC) with the area under the ROC (AUROC) was calculated for the pooled population. Furthermore, the AUROC of time-to-ROSC, a well-established predictor of mortality, was calculated for comparison. To test if PECAM-1 had any additional prognostic value over time-to-ROSC, we investigated a combined AUROC curve with time-to-ROSC and PECAM-1.

Given that neuron-specific enolase (NSE) is currently the only blood-based biomarker recommended for prognostication after cardiac arrest, an exploratory ROC analysis at 48 h was performed to evaluate whether PECAM-1 provided additional prognostic information beyond NSE.[27,28]

We drew boxplots to compare endothelial markers over time in 30-day survivors and non-survivors. For syndecan-1 box plots were made for both the pooled population and in each study group, while PECAM-1 and sTM were only examined in the pooled population. Lastly, for the patients that did not undergo acute CAG at admission, and thus were not heparinized, box plots with individual measurements for syndecan-1, sTM, and PECAM-1 were prepared.

Results

Patient characteristics

In total, 80 patients were included in the IMICA trial, with 39 randomized to receive tocilizumab and 41 assigned to the placebo group (for patient characteristics between groups, we refer to the main study[24]). The median age of the tocilizumab and placebo groups was 65 years (IQR 53–73) and 60 years (IQR 57–70), with a 30-day mortality of 14/39 (36%) and 14/41 (34%), respectively. Median time-to-ROSC in the treatment and placebo groups was comparable, at 20 min (IQR 14–27) and 17 min (IQR 11–28), respectively. Acute CAG was performed for 35 patients in the tocilizumab group and 39 in the placebo group, resulting in markedly elevated baseline activated Partial thromboplastin time (APTT) values, with a median of 241 s in both groups (IQR 28–241 in the tocilizumab and 33–241 in the placebo group). No significant difference was observed between the groups ($P = 0.15$).

Trajectory of endothelial biomarkers

In both groups, syndecan-1 plasma concentrations were generally highest at baseline, followed by a decline over the next 24 h, after which remaining relatively stable through to 72 h (*Figure 1A*). The decline in syndecan-1 concentration was more pronounced for the placebo group at 24, 48, and 72 h (all $P < 0.003$).

The sTM plasma concentrations were generally relative stable over time in both groups (*Figure 1B*). There was no significant

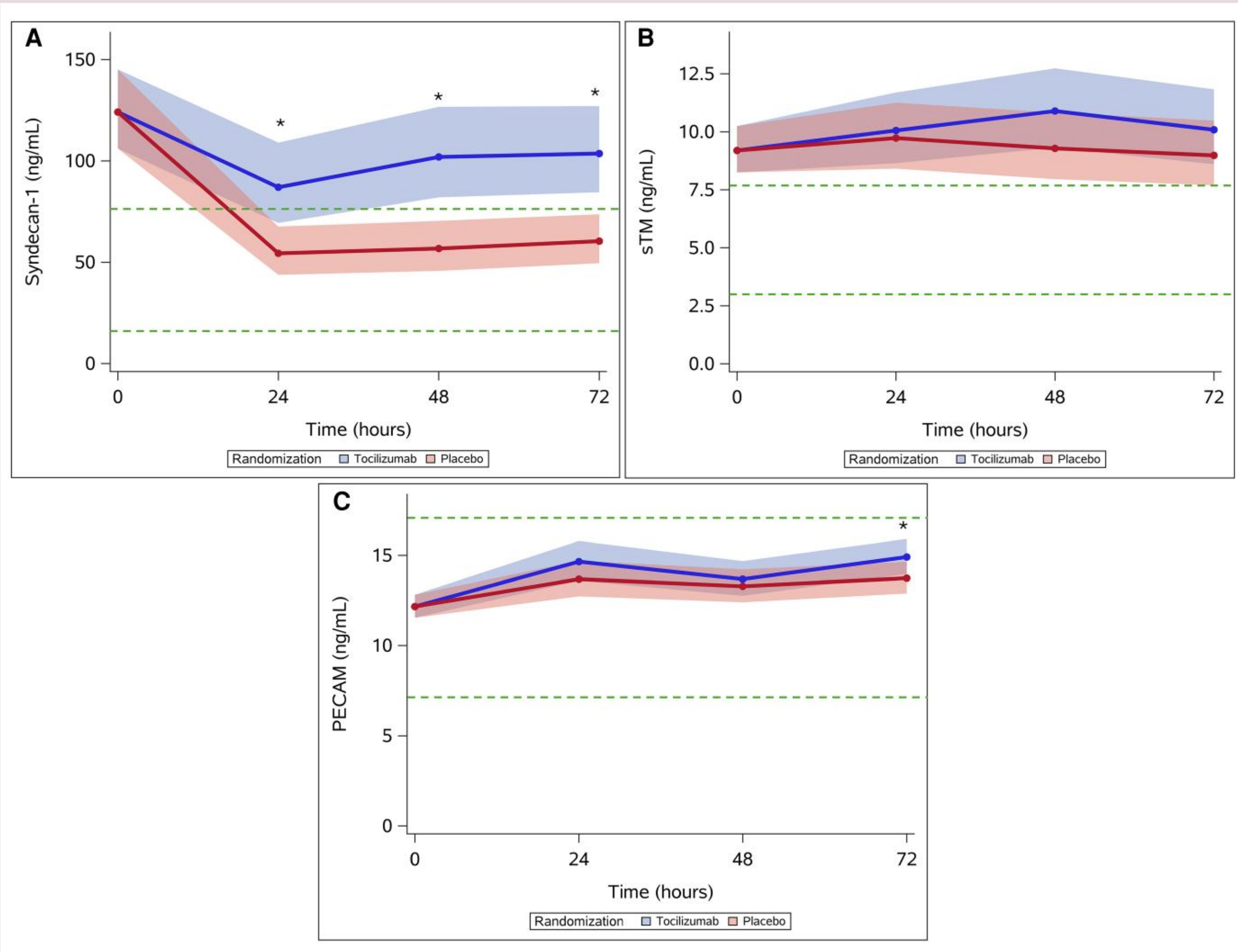


Figure 1 Trajectory of endothelial biomarkers. Results are depicted as predicted means with 95% CI intervals from linear mixed-model analysis. The green dotted line represents the reference interval among healthy individuals. *$P \leq 0.05$ for between-group comparisons. (*A*) Syndecan-1. (*B*) sTM. (*C*) PECAM-1.

difference in sTM plasma concentration between groups during the measurement period.

The trajectory of PECAM-1 generally increased over the course of the 72 h measurement period, exhibiting a biphasic pattern in both groups (*Figure 1C*). Only at the 72 h mark, there was a significant difference in PECAM-1 values, being higher in the tocilizumab group ($P = 0.03$).

The biomarker availability at each time point was generally high with an overall sample availability of 93, 93, and 94%, respectively, for syndecan-1, sTM, and PECAM-1 irrespective of patient status (see Supplementary material online, *Tables S1* and *S2*).

Endothelial biomarkers and 30-day mortality

Since there was a significant difference in syndecan-1 plasma concentration between the two groups at 24–72 h, we assessed syndecan-1 plasma concentration in each group as well as in the pooled population, while PECAM-1 and sTM were only evaluated in the pooled population (see Supplementary material online, *Figures S1–S3*).

For syndecan-1 there was no difference between 30-day survivors and non-survivors at 0–72 h when assessing the cohort as a whole or for the placebo group, while in the tocilizumab group, there were substantially higher levels observed for non-survivors at 0 h ($P = 0.016$) but not at 24–72 h.

sTM plasma concentrations did not differ in non-survivors compared with survivors at 0–72 h. PECAM-1 plasma concentrations were significantly higher among non-survivors at all examined time points (all $P < 0.02$).

The predictive capabilities of baseline endothelial biomarkers on 30-day mortality were analysed by calculating AUROC values (*Figure 2*). Of the three endothelial markers, PECAM-1 performed the best with an AUROC value of 0.71 (95% CI 0.59–0.83), followed by syndecan-1 and sTM, with AUROC values of 0.61 (95% CI 0.48–0.74) and 0.53 (95% CI 0.40–0.67), respectively. The confidence intervals for syndecan-1 and sTM included 0.50, indicating a lack of significant predictive

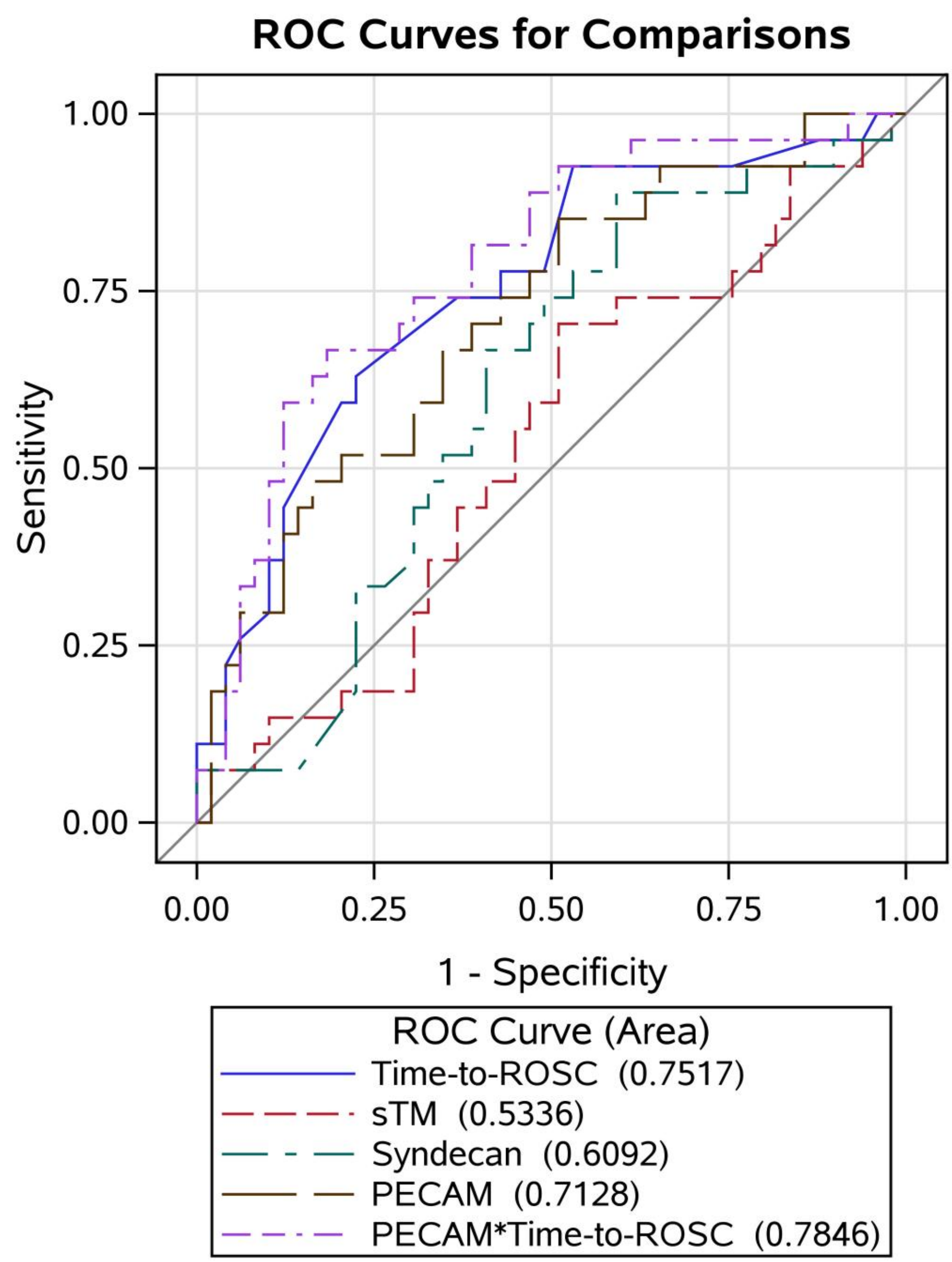


Figure 2 Receiver operating characteristic curves showing baseline endothelial markers and time-to-ROSC predictive value on 30-day mortality.

capability. AUROC for time-to-ROSC was 0.75 (95% CI 0.64–0.87). Time-to-ROSC was significantly better than sTM (P = 0.012), trended towards being better than syndecan-1 (P = 0.079), but not superior to PECAM-1 (P = 0.66), as assessed by the DeLong test. Finally, the combined variable of PECAM-1 and time-to-ROSC revealed the numerically highest AUROC value of 0.78. The combined analysis was not significantly better than time-to-ROSC alone (P = 0.16).

In an exploratory analysis at 48 h, PECAM-1 had an AUROC of 0.72 (95% CI 0.59–0.84) for prediction of 30-day mortality, whereas NSE had an AUROC of 0.92 (95% CI 0.84–0.99). The combination of PECAM-1 and NSE resulted in an AUROC of 0.86 (95% CI 0.75–0.97). A total of 69 patients had both biomarkers available at 48 h.

Discussion

Our findings highlight the endothelial damage and dysfunction that occur in comatose survivors of OHCA, as reflected by the observed syndecan-1 and sTM plasma concentrations that were above the reference interval for healthy individuals. Over the 72 h observation period, there was a less pronounced decrease in syndecan-1 plasma concentration from 24 to 72 h and slightly higher PECAM-1 plasma concentration at 72 h in the active treatment group compared with the placebo group. For the prediction of 30-day mortality based on admission levels, PECAM-1 proved to be a moderate predictor for mortality, while sTM and syndecan-1 had no predictive value.

Syndecan is a transmembrane proteoglycan found on the endothelial surface, serving as a key component of the luminal glycocalyx. During significant injury and inflammation, the extracellular domain of syndecan is enzymatically cleaved.[29] Similarly, sTM is a membrane-bound protein predominantly located on the luminal surface of endothelial cells; however, it is not part of the glycocalyx.[30] An increase in the plasma syndecan-1 concentration indicates damage to the glycocalyx, while damage to the endothelial cells increases plasma concentration of sTM.[29,30]

Syndecan-1 plasma concentrations were elevated and at their highest at baseline in both groups, after which they decreased and reached a plateau. Conversely, sTM was relatively stable over the observed period. Our findings with respect to syndecan-1 align with previous studies that have shown syndecan-1 reaching its maximum value at baseline, while other studies have found that sTM gradually increases in the first days following OHCA.[6,31] These findings suggest that glycocalyx damage may occur soon after resuscitation, while endothelial cell injury might develop gradually over the following days.

Patients treated with tocilizumab had a significantly less pronounced fall in syndecan-1 plasma concentration over the period of 24–72 h compared with placebo. This finding may be explained by three factors: (i) baseline differences between groups, (ii) a biological effect of IL-6 inhibition, or (iii) a chance finding. Despite no group difference in baseline syndecan-1 concentration, differences in comorbidity may still affect the subsequent endothelial response, as patients with higher comorbidity may have a reduced capacity for endothelial repair, leading to a less pronounced decline over time.

Although clinical data on IL-6–mediated endothelial injury following cardiac arrest are limited, recent work by Drost *et al.*[32] showed that IL-6 can directly cause endothelial glycocalyx degradation in patients with COVID-19 and bacterial sepsis. Additionally, they showed that tocilizumab attenuates this effect *in vitro*.[32] An older experimental study has demonstrated that IL-6 can suppress syndecan-1 expression on B-lymphocytes via a rapid post-transcriptional mechanism, which is most pronounced within the first 24 h.[33] Accordingly, we observed a difference between groups after 24 h, remaining constant thereafter. IL-6 inhibitory effect on syndecan-1 may potentially reflect reduced availability for shedding in the acute phase—a mechanism by which the endothelium may protect itself during injury and inflammation.

In our study, renal replacement therapy (RRT) use was more frequent in the tocilizumab group, although this may be a chance finding,[24] RRT has been associated with nearly doubled plasma syndecan-1 concentrations, possibly due to oxidative stress and cytokine release.[34] We found no significant difference in syndecan-1 levels between tocilizumab-treated patients who did and did not receive CRRT (see Supplementary material online, *Figure S4*); however, the analysis was limited by small sample size.

To our knowledge, this is the first study measuring PECAM-1 as an endothelial marker following OHCA. PECAM-1 is highly expressed at the intercellular junctions of endothelial cells and is present on the surface of circulating platelets and leukocytes.[35] Relative to other cellular adhesion molecules previously studied in the context of OHCA, PECAM-1 is located most basally, and an increase in this biomarker would suggest significant endothelial damage.[35] In our study, over the period of 72 h, PECAM-1 seems to exhibit a biphasic trajectory in which the initial rise may be driven by ischaemia–reperfusion-induced endothelial injury with proteolytic shedding. The subsequent decline probably reflects consumption and clearance, and the later increase may correspond to upregulation and re-expression during endothelial repair.[35]

The baseline value of PECAM-1 was superior to the other two endothelial biomarkers in predicting 30-day mortality, with an AUROC value of 0.71. The numerical AUROC value for the combined variable time-to-ROSC × PECAM-1 exceeded that of each marker individually, suggesting that these parameters provide some complementary prognostic information. The predictive abilities of PECAM-1 on mortality following OHCA have not been investigated prior. Our findings contrast with two previous studies, with similar study populations, which have shown that sTM was significantly associated with 30-day mortality [multivariate HR 1.71 (1.05–2.77), $P = 0.031$] and syndecan-1 had a fairly good predictive capability for 28-day mortality (AUROC 0.708).[10,36] Differences in patient characteristics and patient heterogeneity among these studies may explain the differing findings.

Exploratory analyses did not indicate that PECAM-1 adds prognostic information beyond NSE at 48 h.

The moderate predictive ability of PECAM-1 is further underlined by our findings from the comparison of endothelial markers in survivors vs. non-survivors, where PECAM-1 plasma concentration was significantly higher among non-survivors at all time points. Baseline syndecan-1 was significantly higher among non-survivors in patients treated with tocilizumab, while there were no differences observed for the placebo group. In the tocilizumab group, the interquartile range of syndecan-1 plasma concentrations was larger than placebo, which may be attributed to more heterogeneity within this group.

Currently, endothelial markers are not considered when evaluating the prognosis among OHCA patients.[37] Our results suggest that PECAM-1 may serve as a predictor for mortality. However, as this is the first study to evaluate the predictive capabilities of PECAM-1, further research is required to validate these findings. Furthermore, it seems that tocilizumab does not decrease plasma concentrations of endothelial markers over the period of 72 h compared with placebo. Syndecan-1 plasma concentrations were significantly higher in the tocilizumab group, and the responsible mechanism, as well as possible clinical implications, remains unknown. However, previously published results from the IMICA trial demonstrated that early tocilizumab treatment had positive clinical implications, including reduced inflammation, less myocardial injury, and improved haemodynamics with a lower vasoactive–inotropic score at 24 h. These findings are consistent with data from the ASSAIL-MI trial in ST-elevation myocardial infarction (STEMI) patients, where tocilizumab improved myocardial salvage but did not produce clear changes in circulating endothelial or platelet-derived chemokines, suggesting that IL-6 inhibition may exert cardioprotective effects through mechanisms not fully explained by standard endothelial biomarkers.[38]

Limitations

The limitations of this study include its small sample size and single-centre design. Due to the small sample size, some baseline imbalances were present, with patients in the tocilizumab group having a higher prevalence of heart failure and ischaemic heart disease. Furthermore, patients in the placebo group had a higher prevalence of STEMI (59% vs. 33%) and acute percutaneous coronary intervention (54% vs. 34%) both of which may have caused further endothelial damage and thereby influenced the results. Since it was a substudy of the IMICA-trial, it was not originally designed to assess the effects of tocilizumab on endothelial biomarkers and thus was not powered to detect a potential effect. A large proportion of the patients in both study groups underwent acute CAG (89.7% of tocilizumab and 95.1% of placebo patients). At the initiation of this procedure, the patients are systemically heparinized according to national guidelines. Treatment with heparin could potentially have affected the plasma concentration of the measured endothelial biomarkers. Lastly, the study only included OHCA patients with presumed cardiac cause, which limits its external validity.

Conclusion

Tocilizumab infusion attenuated the decline in syndecan-1 plasma concentration observed in the placebo group in the days after cardiac arrest, with no signs of effect on sTM and PECAM-1.

This contrasts with previous findings of the IMICA trial, which showed that tocilizumab treatment led to reduced systemic inflammation, lesser vasopressor usage, and less myocardial injury, suggesting a complex relationship between inflammatory pathways and endothelial injury. Only PECAM-1 showed a moderate association with mortality after resuscitated cardiac arrest.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

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Conflict of interest: C.H. is supported by research grants from the Danish Heart Foundation. J.K. is supported by research grant from the Novo Nordisk Foundation. A.S.P.M.: has been employed outside of this work by Novo Nordisk, Denmark, since 2020.

Data availability

Pending approval from the trial steering committee and relevant authorities; de-identified data can be made available upon reasonable request.

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